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Introduction to Statistical methods (S2-25 AIMLCLZC418) - Assignment 1

AIML - Section 2

1) In a survey it is found that 50% of the men like Tea, 40% like coffee, 30% like juice, 20% like Tea & Cofee, 15% like Coffee & Juice, 10% like tea & juice, 5% like all three. Find the probability that a person likes at least one drink.

A) Step-1: given probabilities for individual drinks and for overlaps (intersections).

We want: $P(\text{tea} \cup \text{Coffe} \cup \text{juice})$

that is, the probability that a person likes at least one of the three drinks.

Let: T = likes tea, C = likes coffe, J = like Juice

Given:

$$P(T) = 0.50, \quad P(C) = 0.40, \quad P(J) = 0.30$$

$$P(T \cap C) = 0.20, \quad P(C \cap J) = 0.15, \quad P(T \cap J) = 0.10$$

$$P(T \cap C \cap J) = 0.05$$

Step-2: ~~Inclusion~~

Inclusion-Exclusion principle for three events for three events A, B, C

$$\boxed{P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)}$$

Step-3 (Apply the formula)

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$P(T \cup C \cup J) = P(T) + P(C) + P(J) - P(T \cap C)$
$\quad - P(C \cap J) - P(T \cap J) + P(T \cap C \cap J)$

$= 0.50 + 0.40 + 0.30 - 0.20 - 0.15 - 0.10 + 0.05$

$= 0.90 + 0.30 - 0.20 - 0.15 - 0.10 + 0.05$

$= 1.20 - 0.20 - 0.15 - 0.10 + 0.05$

$= 1.00 - 0.15 - 0.10 + 0.05$

$= 0.85 - 0.10 + 0.05$

$= 0.75 + 0.05$

P(2)

$P(T \cup C \cup J) = 0.80$

This means there is an 80% chance that a randomly selected person like atleast one of the three drink (Tea, Coffee or Juice)

(2) The probability that a man hits the target is 1/3. if the fire 6 times what is the chance of hitting i) Exactly once ii) at least once iii) at most 5 times.

(A) Given:

probability of hitting the target : $p = 1/3$

probability of missing the target : $q = 1 - p = \frac{2}{3}$

Number of shots : $n = 6$

The binomial probability formula is:

$$P(X = r) = \binom{n}{r} p^r q^{n-r}$$

where X is the number of hits.

(i) probability of hitting exactly once

Here $r = 1$, $P(X=1) = \binom{6}{1}\left(\frac{1}{3}\right)^1\left(\frac{2}{3}\right)^5$

$= 6 \times 1/3 \times 32/243$

$= 64/243$

$P(X=1) = 64/243 \approx 0.2634$

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(ii) (Probability of hitting at least once)

(A) " At least one" means 1 or more hits.
Using the complement rule:

$$P(X \geq 1) = 1 - P(X=0)$$

$$P(X=0) = \left(\frac{2}{3}\right)^6 = \frac{64}{729}$$

therefore, $P(X \geq 1) = 1 - \frac{64}{729} = \frac{665}{729} \approx 0.9122$

(iii) probability of hiting at most 5 times

(A) At most 5 times" means 0, 1, 2, 3, 4 or 5 hits.
using the complement rule:

$$P(X \leq 5) = 1 - P(X=6)$$

$$P(X=6) = \left(\frac{1}{3}\right)^6 = 1/729$$

Hence,

$$P(X \leq 5) = 1 - \frac{1}{729}$$

$$= \frac{728}{729} \approx 0.9986$$

(3.) A university offers scholarships. 50% of applicants are from urban areas, 30% are from rural area. 20% are international students. A randomly selected applicant received a scholarship.

(a) ~~Find the probability that the student is from a rural area.~~ given:

Let: $P(U) = 0.50$ (Urban)
$P(R) = 0.30$ (Rural)
$P(I) = 0.20$ (International)

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Since every applicant belongs to exactly one category :

$$P(U) + P(R) + P(I) = 1$$

A Scholarship recipient is selected at random and no additional information is given about diffrent Scholarship rates. Therefore, the probabilities are the Same as the proportions of applicants.

(a) probability that the Student is from a rural area?
$P(R) = 0.30$ (or) 30%.

(b) probability that the Student is from an urban area
$P(U) = 0.50$ (or) 50%.

(c) which category benefits most ?
Compare the proportions:
Urban = 50% , rural = 30% , International = 20%
The larget proportion is "Urban".

(d) Interpret the fairness of the process.?
A fair process would give equal opportunity to all applicants regardles of background. Here. an urban Student is 2.5 times more likely to get a Scholarship than an international Student (50% vs 20%), and also more likely than a rural Student.

possible reasons for unfairness:
* Bias in selection Criteria
* Different academic preparation levels
* Unequal access to application resource.
* Quata System favoring Urban ares.

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Conclusion: The process is not fair – urban Students benefit disproportionately.

Not fair - Urban Students are overrepresented

(e) If a Scholarship recipient is chosen at random, what is the probability that the Student is either rural or international.

(A) Since the events are mutually exclusive (no Student is both rural and international):

$$P(R \cup I) = P(R) + P(I)$$
$$= 0.30 + 0.20 = 0.50 = 0.50$$

Half of all Scholarship recipients come from either rural areas or international backgrounds.